

CONSTRUCTION AND PROJECT MANAGEMENT I

Section B - Lecture 04: CPM & PERT

Term-Final Question-to-Slide Mapping | 2016-2024

Ex-06/ V. K. Kapoor:
A project has the following characteristics:

Activity	Preceding activity	Expected completion time (weeks)
A	None	5
B	A	2
C	A	6
D	B	12
E	D	10
F	D	9
G	D	5
H	B	9
I	C, E	1
J	G	2
K	F, I, J	3
L	K	9
M	H, G	7
N	M	8

Example 10: A small project consisting of eight activities has the following characteristics:

Activity	Preceding activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	4	12
B	-	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

Deck 4.0 L45: exact 2016 CPM dataset

Deck 4.1 L6: repeated A-H PERT dataset

32

question-to-slide links

479

linked examination marks

Full question marks; partial matches included

8 / 13 / 11

Exact / Direct / Partial

9 years

2016 through 2024

Coverage strength: 357 of 479 linked marks are Exact or Direct matches. The two uploaded decks are strongest for deterministic CPM calculations, float definitions and PERT numerical procedures. Original Section A/B labels are retained even where yearly section order changed.

Finding	Evidence
Exact 2016 repeat	The complete 2016 A2(c) A-N activity table and four tasks are printed on Deck 4.0 L45-46.
Three-time repeated PERT dataset	The A-H dataset used in 2017 A4(c), 2019 A4(b) and 2024 A4(b) is Deck 4.1 L6-9.
Most repeated theory block	Float, slack, critical path and ES/EF/LS/LF definitions appear in seven examination years.
Main limitations	GERT and PERT-specific advantages/shortcomings are not sufficiently covered in these files.

1. Lecture Coverage Map and Repetition Pattern

Deck / slides	Core content	Question-bank use
4.0 L1-7	Project scheduling purposes, techniques and Gantt charts	2016 B8(a); supporting context for scheduling questions
4.0 L8-10	CPM/PERT overview and six-step method	CPM-vs-PERT, network-method and procedure questions
4.0 L11-15	Critical path, slack, forward/backward pass, ES/EF/LS/LF and total float	Definition questions and all deterministic calculations
4.0 L16-42	Milwaukee Paper complete forward/backward-pass example	Transferable method for 2017, 2020, 2021 and 2023 CPM problems
4.0 L33-39	Total, free, independent and AOA/AON floats	Repeated theory in 2017, 2018, 2019, 2020, 2023 and 2024
4.0 L43-49	Exercises, exact 2016 A-N dataset and additional PERT-style case	Network analysis and exam practice
4.0 L51-53	Advantages and limitations of CPM	Exact 2016 A2(b) match
4.1 L1-5	PERT expected time, variance, network and normal probability example	Direct support for probability-based PERT questions
4.1 L6-9	A-H PERT dataset, schedule and network	Exact dataset repeat in 2017, 2019 and 2024

Question family	Years / frequency	Primary slides	Study implication
PERT numerical analysis	2016-2019 and 2021-2024 (8 years)	4.1 L1-9 plus 4.0 L12-39	Highest priority; practise networks, expected time, variance, float and probability.
Deterministic CPM numerical analysis	2016, 2017, 2020, 2021, 2023 (5 years)	4.0 L11-49	Master forward/backward pass and sensitivity to duration change.
Float / slack / critical-path definitions	2016, 2017, 2018, 2019, 2020, 2023, 2024 (7 years)	4.0 L11-14 and L33-39	A reliable theory question and essential numerical foundation.
CPM advantages / limitations	2016	4.0 L51-53	Small but exact and easy to secure.
Gantt / PERT / GERT mixed theory	2016	4.0 L2-8	Only partly covered; add separate GERT notes.
Network drawing without calculations	2019, 2020, 2023, 2024	4.0 L9-10 and L43-49	Use Lecture 03 for formal AOA/AON and dummy rules.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024
Linked marks	85	70	35	50	42	40	33	69	55

Important source correction: Deck 4.1 L6 gives activity B pessimistic time as 26, matching the examinations. L8 prints 25 in the solution table, but its expected time 14 and variance 64/9 are mathematically consistent with 26. Use 26.

2.1 Detailed Question-to-Slide Mapping: 2016-2018

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2016	A 2(a)	4	1	Define slack time and critical path.	4.0 L11-14, L33-35	Exact	Both terms, zero-float condition and critical-path meaning are stated directly.
2016	A 2(b)	6	1	Write the advantages and limitations of CPM.	4.0 L51-53	Exact	The final three slides list CPM advantages and limitations in exam-ready form.
2016	A 2(c)	25	1	AOA CPM problem: paths, critical path, completion time, ES/EF/LS/LF/float, and changed duration.	4.0 L12-46	Exact	The exact A-N activity dataset and all four requested tasks appear on L45-46; L12-42 gives the calculation method.
2016	A 3(a)	10	2	Distinguish between CPM and PERT.	4.0 L8-10	Partial	The deck identifies both as network techniques and notes their different time estimates, but does not provide a full comparison table.
2016	A 3(b)	25	2	Analyze a twelve-activity project using PERT; calculate floats, critical path and 99% completion time.	4.1 L1-9; 4.0 L12-39	Direct	PERT formulas, probability workflow and scheduling method are covered with different worked datasets.
2016	B 8(a)	15	3	Define PERT and GERT; state Gantt-chart objectives and preparation method.	4.0 L2-8	Partial	Scheduling purposes, PERT overview and two Gantt examples are present; GERT is not explained.
2017	A 2(a)	15	4	Define total, free and independent float, critical path, early finish and late start.	4.0 L11-14, L33-39	Exact	All requested terms and formulas are explicitly provided.
2017	A 2(b)	20	4	Draw the network, prepare ES/EF/LS/LF/float, identify critical path and test a duration change.	4.0 L12-44	Direct	The complete deterministic CPM workflow is taught with comparable networks and worked calculations.
2017	A 4(a)	8	5	Define optimistic, most likely and pessimistic times and variance.	4.1 L1-3	Exact	The three estimates and variance formula appear together.
2017	A 4(b)	7	5	Write the advantages and shortcomings of PERT.	4.0 L8, L51-53	Partial	Only general network-method context and CPM-specific advantages/limitations are supplied; PERT-specific shortcomings are absent.
2017	A 4(c)	20	5	Eight-activity PERT problem: network, critical path and 99% confidence before a 30-week deadline.	4.1 L6-9	Direct	The exact A-H dataset and requested tasks are reproduced; the probability arithmetic is not fully displayed.
2018	B 8(a)	10	8	Short notes on CPM, free float, variance, optimistic time and pessimistic time.	4.0 L8-14, L33-39; 4.1 L2-3	Exact	Every requested concept is directly covered across the two decks.
2018	B 8(b)	5	8	Write the advantages and shortcomings of PERT.	4.0 L8, L51-53	Partial	PERT-specific advantages and shortcomings are not stated; the closest material is CPM/network-method evaluation.
2018	B 8(c)	20	8	Analyze a twelve-activity project by PERT and determine total float, critical path and 99% completion time.	4.1 L1-9; 4.0 L12-39	Direct	The lecture teaches the same PERT and float procedure, but uses different example data.

QB = original question-bank PDF page. 4.0 and 4.1 refer to the two uploaded Lecture 04 PDF files.

2.2 Detailed Question-to-Slide Mapping: 2019-2021

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2019	A 3(c)	15	9	Infer activity relationships and draw a network for a steelwork project.	4.0 L9-10, L43-49	Partial	Network construction is an input to CPM and comparable exercises are shown, but relationship inference is treated more fully in Lecture 03.
2019	A 4(a)	10	9	Define free float, independent float, dummy activity, optimistic time and pessimistic time.	4.0 L37-39; 4.1 L1-3	Partial	Float and PERT-time terms are covered; a formal dummy-activity definition is not included.
2019	A 4(b)	25	10	Eight-activity PERT problem: network, critical path, activity schedule, all floats and 99% start time.	4.1 L6-9; 4.0 L33-39	Direct	The exact A-H dataset is reproduced and float formulas are available; the final probability/start-time arithmetic must be completed.
2020	A 2(a)	10	12	Define critical float, slack time, PERT, free float and independent float.	4.0 L8, L11-14, L33-39	Direct	All ideas are covered, although the deck uses zero total float on the critical path rather than the phrase critical float.
2020	A 2(b)	20	12	Draw the network and calculate critical path, total float, true/free float and independent float.	4.0 L12-49	Direct	Forward/backward pass and all requested float types are explained with worked examples.
2020	A 3(c)	12	13	Draw an AOA network for a construction project containing dummy activities.	4.0 L9-10, L43-49	Partial	AOA-style examples and zero-duration links appear, but formal AOA/dummy construction rules belong to Lecture 03.
2021	A 3(c)	12	15	Draw the AOA network and mark the critical path.	4.0 L9-14, L40-46	Direct	Network-to-critical-path analysis is central to the deck; the exact dataset is different.
2021	A 4(c)	20	16	Analyze a twelve-activity project using PERT and find total float, critical path and 99% completion time.	4.1 L1-9; 4.0 L12-39	Direct	The same analysis procedure is covered using two worked PERT examples.
2021	B 8(b)	8	17	Define GERT and LSM and differentiate GERT from CPM or PERT.	4.0 L8	Partial	Only CPM/PERT background is present; GERT and LSM definitions are outside these decks.

QB = original question-bank PDF page. 4.0 and 4.1 refer to the two uploaded Lecture 04 PDF files.

2.3 Detailed Question-to-Slide Mapping: 2022-2024

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2022	A 4(a)	25	18	Analyze a twelve-activity project using PERT and determine floats, critical path and 99% completion time.	4.1 L1-9; 4.0 L12-39	Direct	The PERT probability and scheduling workflow is directly transferable to the exam dataset.
2022	B 8(b)	8	19	Define GERT and differentiate GERT from PERT.	4.0 L8	Partial	PERT is introduced, but GERT is not covered and a full comparison cannot be written from these slides alone.
2023	A 2(b)	12	20	Write short notes on WBS, OBS, AOA networks and project planning.	4.0 L9-10, L43-49	Partial	Only the AOA/network component links to this lecture; WBS, OBS and planning are mainly Lecture 03.
2023	A 3(b)	22	20	AON CPM problem: paths, critical path, duration, ES/EF/LS/LF/float and changed duration.	4.0 L11-46	Direct	The full CPM calculation sequence is provided with a comparable AON worked example.
2023	A 4(a)	25	20-21	Analyze an eleven-activity project using PERT; determine floats, critical path and 99% completion time.	4.1 L1-9; 4.0 L12-39	Direct	The same formulas and probability method apply, with different project data.
2023	A 4(b)	10	21	Define total float, free float, independent float, early start and late start.	4.0 L12-14, L33-39	Exact	All requested definitions and formulas are explicit.
2024	A 3(b)	20	23	Draw AOA and AON network diagrams from predecessor information.	4.0 L9-10, L43-49	Partial	The CPM deck relies on networks and shows examples, but AOA/AON drawing rules are more complete in Lecture 03.
2024	A 4(a)	10	23	Define critical path, free float, independent float, variance and slack time.	4.0 L11-14, L33-39; 4.1 L2-3	Exact	Every term and required equation is present.
2024	A 4(b)	25	23-24	Eight-activity PERT problem: network, critical path, schedule and 99% start time.	4.1 L6-9	Direct	The exact A-H dataset is reproduced; the slides stop before the full probability calculation.

QB = original question-bank PDF page. 4.0 and 4.1 refer to the two uploaded Lecture 04 PDF files.

3. Slide-to-Question Reverse Index

Slides	Core topic	Mapped questions	Priority
4.0 L1-7	Scheduling purpose, techniques and Gantt charts	2016 B8(a)	Secondary / mixed
4.0 L8-10	CPM/PERT overview and six steps	2016 A3(a), B8(a); 2019 A3(c); 2020 A3(c); 2021 A3(c), B8(b); 2022 B8(b); 2023 A2(b); 2024 A3(b)	High support
4.0 L11-15	Critical path, slack, forward/backward pass, ES/EF/LS/LF, total float	2016 A2(a-c), A3(b); 2017 A2(a-b); 2018 B8(a,c); 2020 A2(a-b); 2021 A3(c), A4(c); 2022 A4(a); 2023 A3(b), A4(a-b); 2024 A4(a)	Highest
4.0 L16-32	Complete Milwaukee forward/backward-pass calculation	2016 A2(c); 2017 A2(b); 2020 A2(b); 2021 A3(c); 2023 A3(b)	Very high numerical
4.0 L33-39	Slack; total, free and independent float; AOA/AON float formulas	2016 A2(a-c), A3(b); 2017 A2(a-b); 2018 B8(a,c); 2019 A4(a-b); 2020 A2(a-b); 2021 A4(c); 2022 A4(a); 2023 A3(b), A4(a-b); 2024 A4(a)	Highest theory + numerical
4.0 L40-44	Critical path display and practice networks	2017 A2(b); 2020 A2(b); 2021 A3(c); 2023 A3(b)	Practice
4.0 L45-46	Exact A-N CPM exercise	2016 A2(c)	Exact exam reproduction
4.0 L47-49	Additional stochastic/float exercises	2016 A3(b); 2018 B8(c); 2019 A4(b); 2021 A4(c); 2022 A4(a); 2023 A4(a)	Advanced practice
4.0 L51-53	CPM advantages and limitations	2016 A2(b); partial support for 2017 A4(b) and 2018 B8(b)	Exact CPM theory
4.1 L1-5	PERT expected time, variance, network and probability	2016 A3(b); 2017 A4(a,c); 2018 B8(a,c); 2019 A4(a-b); 2021 A4(c); 2022 A4(a); 2023 A4(a); 2024 A4(a-b)	Highest PERT
4.1 L6-9	Exact repeated A-H PERT dataset	2017 A4(c); 2019 A4(b); 2024 A4(b)	Exact dataset, 3 years

Fastest revision route: 4.0 L11-39 -> 4.1 L1-9 -> 4.0 L45-46 -> 4.0 L51-53 -> 4.0 L1-10 -> 4.0 L40-49.

4. Exact Repeats, Verification and Added Corrections

4.1 Exact 2016 CPM question

Exam: 2016 A2(c). **Slides:** 4.0 L45-46. The activity table and all four requested tasks are reproduced verbatim in substance.

Verified baseline result: Critical path A-B-D-E-I-K-L; project duration = 42 weeks.

If G changes from 5 to 10 weeks: the controlling branch changes to A-B-D-G-M-N; new duration = 44 weeks.

Ex-o6/ V. K. Kapoor:
A project has the following characteristics:

Activity	Preceding activity	Expected completion time (weeks)
A	None	5
B	A	2
C	A	6
D	B	12
E	D	10
F	D	9
G	D	5
H	B	9
I	C, E	1
J	G	2
K	F, I, J	3
L	K	9
M	H, G	7
N	M	8

4.2 Three-time repeated A-H PERT dataset

Exams: 2017 A4(c), 2019 A4(b), 2024 A4(b). **Slides:** 4.1 L6-9.

Verified expected durations: A=5, B=14, C=9, D=15, E=8, F=9, G=4, H=5 weeks.

Critical path: A-D-G-H = 29 weeks. Critical-path variance = 6; standard deviation = $\sqrt{6}$ = 2.449 weeks.

99% completion duration: $29 + 2.33 \times 2.449$ = about 34.7 weeks. Relative to a 30-week baseline, start about 4.7 weeks earlier.

Activity	Time estimates					Earliest		Latest	
	a	m	b	t_e	σ^2	Start	Finish	Start	Finish
A	2	4	12	5	25/9	0	5	0	5
B	10	12	25	14	64/9	0	14	1	15
C	8	9	10	9	1/9	5	14	6	15
D	10	15	20	15	25/9	5	20	5	20
E	7	7.5	11	8	4/9	5	13	16	24
F	9	9	9	9	0	14	23	15	24
G	3	3.5	7	4	4/9	20	24	20	24
H	5	5	5	5	0	24	29	24	29

Correction note for Deck 4.1: L8 shows B pessimistic time as 25, while L6 and all three exam papers use 26. The displayed expected time 14 and variance 64/9 confirm that 26 is the intended value. This report standardizes the value to 26 without altering the other lecture content.

5. Exam-Focused Answer Construction Guide and Formula Sheet

Item	Exam-ready formula / rule
Forward pass	$EF = ES + \text{duration}$; ES = maximum EF of all immediate predecessors.
Backward pass	$LS = LF - \text{duration}$; LF = minimum LS of all immediate successors.
Total float	$TF = LS - ES = LF - EF$. Critical activity: $TF = 0$.
Free float	FF = earliest start of successor - earliest finish of current activity. Use the minimum successor ES when there are several successors.
Independent float	$IF = FF - \text{tail slack}$; negative IF is taken as zero.
PERT expected time	$t_e = (a + 4m + b) / 6$.
PERT variance	$\text{Variance} = ((b - a) / 6)^2$.
Project probability	$Z = (\text{target duration} - \text{expected critical-path duration}) / \sqrt{\text{sum of critical-path variances}}$.
Critical path	Longest-duration path through the network; it defines the minimum project completion time.

1. Definitions question

Start with the exact term, then one formula and one interpretation. For float questions, state the hierarchy $IF \leq FF \leq TF$ and $TF=0$ on the critical path.

2. Deterministic CPM numerical

Draw the network, list paths, conduct forward pass, conduct backward pass, calculate floats, identify zero-float chain, then test any changed duration by repeating only affected branches.

3. PERT numerical

Compute t_e and variance for every activity, draw the network using t_e , run CPM on expected times, sum variances on the critical path, then use the normal Z calculation.

4. CPM advantages/limitations

Use 4.0 L51-53. Keep advantages and limitations separate and include the risk of excessive attention to only the current critical path.

5. Gantt-chart mixed question

Use 4.0 L2-7 for scheduling purpose and chart construction, but add a separate GERT definition from another source.

6. CPM vs PERT

Use 4.0 L8 as the shared foundation. Add deterministic single-time estimates for CPM versus probabilistic three-time estimates for PERT; the deck alone is not enough for a full 10-mark comparison.

7. Exact 2016 A-N dataset

Study 4.0 L45-46 first. Memorize the verified baseline CP and understand how increasing G redirects the controlling branch.

8. Repeated A-H PERT dataset

Study 4.1 L6-9 and the correction note. Practise the 29-week CP and 34.7-week 99% completion duration.

Formula correction: Deck 4.0 L37 prints the free-float subtraction in reverse order. Use $FF = \text{minimum } ES \text{ of immediate successors} - EF \text{ of the current activity}$. For independent float, use $IF = FF - \text{tail slack}$ and take a negative result as zero.

6. Coverage Limits and Final Study Priority

Question area	Support in Lecture 04	Action required
GERT definitions and comparisons	Not covered beyond mentioning CPM/PERT as network methods	Use separate GERT notes; do not construct a full answer from these slides.
PERT advantages and shortcomings	No PERT-specific list; L51-53 is CPM-specific	Use another lecture/reference for a complete 5-7 mark response.
AOA/AON construction and dummy rules	Examples are present, but formal drawing rules are limited	Use Section B Lecture 03 slides L47-60 as the primary source.
Probability table reading	4.1 includes one normal-distribution lookup example	Practise Z-table direction and one-tailed interpretation independently.
Exact solutions for every exam dataset	Methods and two major datasets are supplied	Fresh datasets still require complete calculations during practice.
Terminology inconsistencies	Critical float / true float wording varies by paper	Answer using standard total/free/independent float definitions and clarify the examiner term.

Study order	Slides	Focus
Priority 1	4.0 L11-39	Forward/backward pass, critical path, ES/EF/LS/LF and all float formulas
Priority 2	4.1 L1-9	PERT expected time, variance, probability and repeated A-H dataset
Priority 3	4.0 L45-46	Exact 2016 A-N CPM question and duration-change analysis
Priority 4	4.0 L40-49	Additional CPM/PERT network practice
Priority 5	4.0 L51-53	Exact CPM advantages and limitations
Priority 6	4.0 L1-10	Scheduling/Gantt overview and CPM/PERT procedure

Bottom line: Lecture 04 is one of the highest-yield blocks in the question bank. Master 4.0 L11-39 and 4.1 L1-9 first, then practise the exact repeat datasets. Keep separate notes for GERT and PERT-specific advantages/shortcomings.